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Methods of Planning Analysis

Prof. Powers

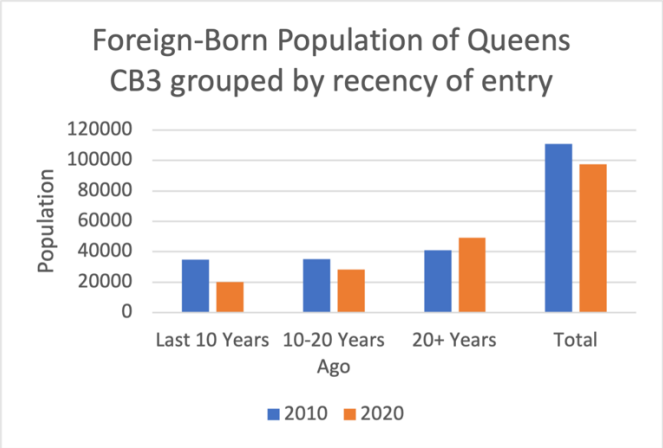
Assignment 3

From 2010 to 2020, Queens Community Board Three's foreign-born population underwent modest changes. Among these changes are slight decreases to the foreign-born population, slight increases to mean household earnings, higher levels of rent burden and a decrease in the percentage of the population that is below the poverty level. From this data, we can observe that although conditions improved for the foreign-born population living in Community Board 3, it is likely that the factors of gentrification are present, which is evident by a declining population, higher rent-burdens and higher average income.

When looking at the population of the district, we can see that foreign-born population had decreased across many variables. Firstly, we can confidently say that the difference in population from 2010 to 2020 is statistically significant, as seen below in **Significance Test #1**, where the significance of the change in the variable is calculated. Therefore, we conclude that the population of the district did decrease over the ten-year period, by 13,420 foreign-born residents, a 12.4% decrease. This differs from general population trends in New York City over the same time period, where the city gained about 629,000 residents, or 7% (Kober). The annual rate of population growth for the city was 0.74% per year, while the district had a foreign-born annual rate of decline of 1.28% per year. Clearly, there is a trend that despite population growth in the city, there is a decline in the foreign-born population in Queens CB3.

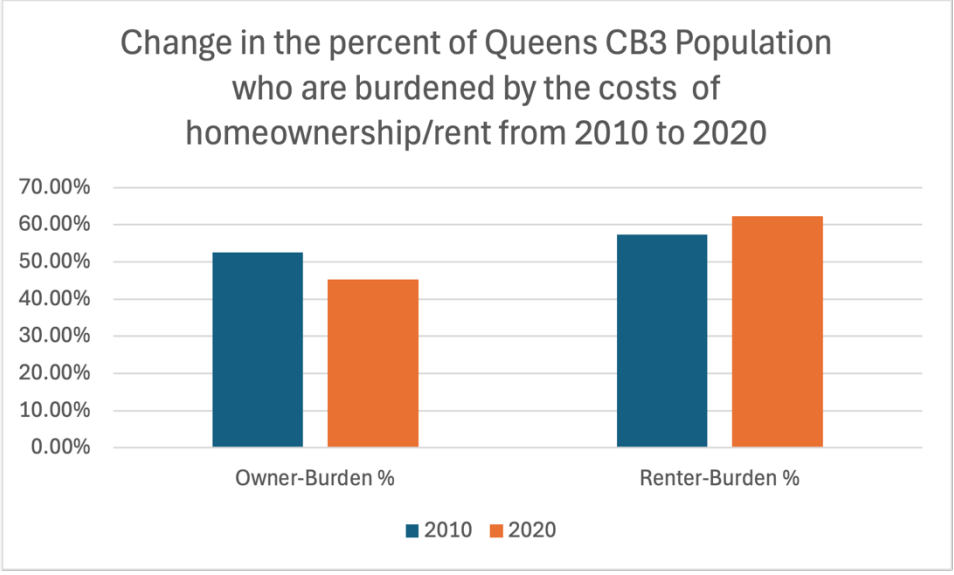
Significance Test #1

Category	Total Foreign-Born Population
90% MOE Z-score	1.645
2020 Value	97,474 (MOE 3,899)
2010 Value	110,894 (MOE 2,508)
X_1-X_2	13420
SE_1	2,370
SE_2	1,525
Denom	Sqrt($2370^2 + 1525^2$) = 2818
Significant?	**YES**, 13420/2818 is greater than Z(1.645)



Contributing to the population decline, in all categories except for foreign-born residents who have been in the US for more than twenty years, the population in 2020 lagged behind the population from 2010. When considering the effects of neighborhood change, such as gentrification, this data makes plausible sense. As the neighborhood gentrifies, the growth of population of new foreign-born entries declines since the neighborhood is now more expensive, and more US-born residents move in. Only the long-time residents can remain since they have much higher rates of home ownership compared to those who haven't been in the US for as long. In 2020, more than 40% of residents who entered over twenty years ago are homeowners. That is more than double the rate of residents who enter 10-20 years ago (17% homeownership) and more than five times the rate of residents who entered within the last 10 years (7% homeownership). Compared to 2010, there is a 5.2% decrease in owner-occupied homes among longtime homeowners, so it is possible that despite higher home ownership, they are still feeling the effects of neighborhood change. With some indicators of gentrification present, we will look to see if the community is burdened by changing costs of living which would likely also be present if the neighborhood was experiencing gentrification.

To determine the full extent of housing burden, we will look at both the rate of homeowners whose housing costs exceed 30% of their household income, and the rate of renters whose rent exceeds 30% of their household income. In 2010, that rate was 52.5% for homeowners, and 57.4% for renters. This is significant, as more than half of both categories are burdened with high costs of housing. In 2020, this rate changes in a very interesting way. It declines 7.2 percentage points to 45.3% for homeowners, while at the same time increasing 4.9 percentage points to 62.3% for renters. This can also be seen in the chart below. This seems to indicate that the homeowners in the neighborhood are doing significantly better in 2020 (13% less burdened) than they were in 2010, despite renters doing significantly worse (8.5% more burdened).



Although we don't have any data about the income of renters versus homeowners, according to the Federal Reserve, the average homeowner earns more income than the average renter (Choi). This would reinforce the argument that gentrification is occurring in CB3, as wealthier homeowners are becoming less burdened with housing costs, despite their less-wealthy renter counterparts becoming ever-increasingly burdened with their housing costs. To further investigate this hypothesis of gentrification, we will analyze different income variables for the community.

When looking at income, the primary variable is often Median Household Income. In the case of Queens CB3, we can see that there was not a significant change in this variable after adjusting for inflation (Significance Test #2). However, when looking at the average earnings of households with earnings, we do find a statistically significant difference in 2020 compared to 2010, even after adjusting for inflation (Significance Test 2). Therefore, we can conclude that despite MHI not changing, mean earnings did increase for the neighborhood. This is an interesting difference and certainly raises some questions. These results could be consistent with gentrification, where the top earners are making significantly more than ten years ago despite the median remaining mostly unchanged. This may indicate that the foreign-born population is under more financial pressure in 2020 than in 2010.

Significance Test #2

Category	Median Household Income	Mean Earnings (For households with earnings)
90% MOE Z-score	1.645	1.645
2020 Value	$57,311 (MOE $2,871)	$77,038 (MOE $3,109)
2010 Value	$47,632 (MOE $1,752)	$60,599 (MOE $1,836)
2010 Value (Inflation Adjusted)	$56,707 (MOE $2,085)	$72,144 (MOE $2,185)
X_1-X_2	$604	$4,894
SE_1	$1,745	$1,890
SE_2	$1,267	$1,328
Denom	Sqrt($1267^2 + 1745^2$) = 2,156	Sqrt($1890^2 + 1328^2$) = 2,310
Statistically Significant?	**NO**, 604/2156 is not greater than Z(1.645)	**YES**, 4894/2310 = 2.12, which IS greater than Z (1.645)

In addition to just purely income, we also look at the sources of income across the population. Specifically, we will look at social security and retirement-related income to better understand if the theory about long-time homeowner residents remaining despite gentrification

pressures is true. From 2010 to 2020, the number of residents receiving social security increased from 18.2% to 25.2%, and the number receiving retirement income increased from 8.1% to 11.4%.

We also see that the median age of the neighborhood rose from 40 to 45 over the ten-year period. This would mean an annual absolute increase in age of 0.5 years old per year. Although this is slower than what the rate would be if there were no new residents or births (since everyone would age one year old per year), it is still indicative of the overall population aging in a significant manner. By combining the increasing median age with the source of income change, we strengthen the argument that when faced with gentrification, the foreign-born population that can remain is increasingly the older homeowners.

Finally, we can see that there was a decrease in the percent of individuals below the poverty line from about 18% to 13%. On one hand, this could indicate that the general foreign-born population is doing better in 2020 than in 2010. However, when combined with population loss, rising rent burden, lower homeownership and stagnant incomes, we may infer something differently; It is likely that the poorest members of CB3 were no longer able to afford to live in the neighborhood with the increasing pressures of gentrification and were forced to relocate to new places due to the pressure.

Overall, Queens Community Board 3 has likely experienced gentrification that has resulted in a significant decrease of their foreign-born population. Some of the population, particularly homeowners, may be doing better, but at the same time many residents are facing significantly increasing financial pressure if they remain in the neighborhood.

Bibliography

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